MACHINE LEARNING FUNDAMENTALS - LS2026
SEMINAR: PREDICTOR EVALUATION

CZECH TECHNICAL UNIVERSITY IN PRAGUE
V. FRANC

Assignment 1. We are given a prediction strategy $h\colon \mathcal{X} \to \mathcal{Y} = \{1, \ldots, Y\}$ assigning observations $x \in \mathcal{X}$ into one of Y classes. Our task is to estimate the true error $R(p, h) = \mathbb{E}_{(x,y)\sim p}\ell(y, h(x))$ where $\ell\colon \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ is a chosen loss function. To this end, we collect a test set $S_n = ((x_i, y_i) \in (\mathcal{X} \times \mathcal{Y}) \mid i = 1, \ldots, n)$ i.i.d. drawn from the distribution $p(x, y)$, compute the test error $\hat{R}(S_n, h) = \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, h(x_i))$ and use it to construct the confidence interval such that

$$R(p, h) \in \Big(\hat{R}(S_n, h) - \varepsilon, \hat{R}(S_n, h) + \varepsilon\Big) \quad \text{holds with probability } 1-\delta \in (0, 1) \text{ at least.} \tag{1}$$

The number of test examples $n \in \mathbb{N}$, the error margin $\varepsilon > 0$ and the confidence level $1 - \delta \in (0, 1)$ are three interdependent variables, i.e., fixing two of the variables allows to compute the third one.

a) Use the Hoeffding's inequality to derive a formula to compute ε as a function of n and δ such that (1) holds.

b) Use the Hoeffding's inequality to derive a formula to compute n as a function of ε and δ such that (1) holds.

c) Instantiate the formulas derived in a) and b) for the following loss functions:

(1) $\ell(y, y') = [\![y \neq y']\!]$
(2) $\ell(y, y') = |y - y'|$
(3) $\ell(y, y') = [\![|y - y'| \geq K]\!]$ where $K < Y$.

d) Assume that we use the loss $\ell(y, y') = [\![y \neq y']\!]$. Plot the error margin ε as a function of the number of examples $n \in \{10, 100, \ldots, 100000\}$ for $\delta \in \{0.1, 0.05, 0.01\}$.

e) Assume that we use the 0/1-loss $\ell(y, y') = [\![y \neq y']\!]$. What is the minimal number of examples n we need to use to have a guarantee that the test error will approximate the generalization error $\pm 1\%$ with probability 95% at least?

Solution 1. a) Formula for $\varepsilon(n, \delta)$. Assume the loss values are bounded in an interval $[\ell_{\min}, \ell_{\max}]$. Let $Z_i = \ell(y_i, h(x_i)) \in [\ell_{\min} - \ell_{\max}]$, $\hat{R} = \frac{1}{n}\sum_{i=1}^{n} Z_i$ and $R = \mathbb{E}_{S_n\sim p^n}[\ell(y, h(x)]$. Hoeffding's inequality says

$$\mathbb{P}\big(|\hat{R} - R| \geq \varepsilon\big) \leq 2\exp\Big(-\frac{2n\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}\Big).$$

We want $\mathbb{P}(|\hat{R} - R| < \varepsilon) \geq 1 - \delta$ which is the same as $\mathbb{P}(|\hat{R} - R| \geq \varepsilon) \leq \delta$. So, equivalently, we want the right-hand side of the Hoeffding inequality to be $\leq \delta$. Solve for ε:

$$2\exp\Big(-\frac{2n\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}\Big) \leq \delta \quad \iff \quad -\frac{2n\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2} \leq \ln\frac{\delta}{2}$$

which gives

$$\boxed{\varepsilon(n, \delta) = (\ell_{\max} - \ell_{\min})\sqrt{\frac{1}{2n}\ln\big(\tfrac{2}{\delta}\big)}}.$$

With this ε we have $\mathbb{P}\big(|\hat{R} - R| \leq \varepsilon\big) \geq 1 - \delta$.

b) Formula for $n(\varepsilon, \delta)$. Rearrange the same bound to solve for n:

$$2\exp\Big(-\frac{2n\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}\Big) \leq \delta \quad \Longrightarrow \quad n \geq \frac{(\ell_{\max} - \ell_{\min})^2}{2\varepsilon^2}\ln\big(\tfrac{2}{\delta}\big).$$

So one can choose

$$\boxed{n(\varepsilon, \delta) = \frac{(\ell_{\max} - \ell_{\min})^2}{2\varepsilon^2}\ln\big(\tfrac{2}{\delta}\big)}$$

(or the ceiling of the right-hand side to get an integer).

c) Instantiate for the three losses:

(1) $\ell(y, y') = [\![y \neq y']\!]$. This loss takes values in $\{0, 1\}$, so $\ell_{\min} = 0, \ell_{\max} = 1$ and $\ell_{\max} - \ell_{\min} = 1$. Thus

$$\varepsilon = \sqrt{\frac{1}{2n}\ln\big(\tfrac{2}{\delta}\big)}, \qquad n = \frac{1}{2\varepsilon^2}\ln\big(\tfrac{2}{\delta}\big).$$

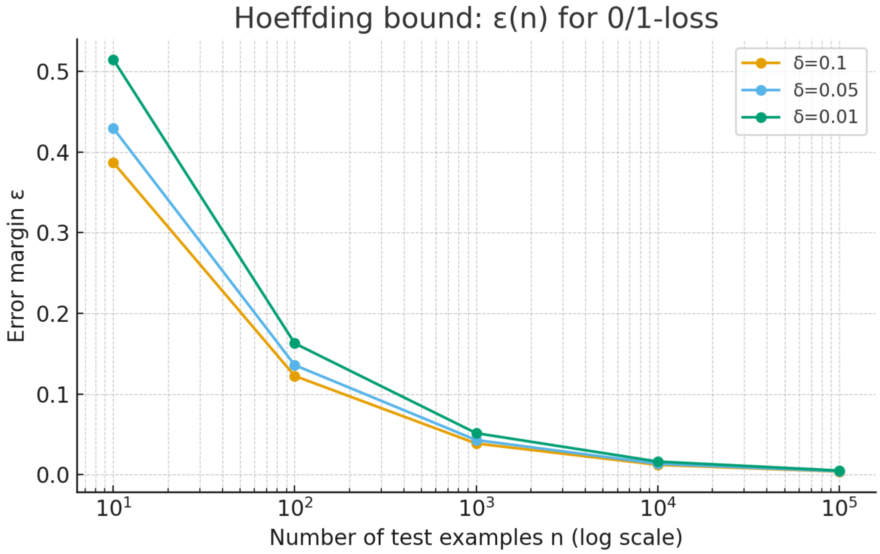
(2) $\ell(y, y') = |y - y'|$ with $y, y' \in \{1, \dots, Y\}$. The range is $[0, Y - 1]$, so $\ell_{\max} - \ell_{\min} = Y - 1$. Hence

$$\varepsilon = (Y - 1)\sqrt{\frac{1}{2n}\ln\big(\tfrac{2}{\delta}\big)}, \qquad n = \frac{(Y - 1)^2}{2\varepsilon^2}\ln\big(\tfrac{2}{\delta}\big).$$

(3) $\ell(y, y') = [\![|y - y'| \geq K]\!]$. This is again a $\{0, 1\}$-valued loss, so the formulas are the same as in 1):

$$\varepsilon = \sqrt{\frac{1}{2n}\ln\big(\tfrac{2}{\delta}\big)}, \qquad n = \frac{1}{2\varepsilon^2}\ln\big(\tfrac{2}{\delta}\big).$$

d) Plot ε vs n for 0/1-loss and $\delta \in \{0.1, 0.05, 0.01\}$.



The curve shows the $O(1/\sqrt{n})$ decay and the dependence on δ via $O(\sqrt{\ln(2/\delta)})$.

e) Minimal n for 0/1-loss, $\varepsilon = 1\%$, $1-\delta = 95\%$. We need $\varepsilon = 0.01$ and $\delta = 0.05$. Use $n \geq \frac{1}{2\varepsilon^2}\ln\left(\frac{2}{\delta}\right)$. Numerically:

$$n \geq \frac{1}{2 \cdot 0.01^2}\ln\left(\frac{2}{0.05}\right) = \frac{1}{2 \cdot 10^{-4}}\ln(40) = 5000 \cdot \ln(40) \approx 18444.3973.$$

Rounding up to an integer,

$$\boxed{n_{\min} = 18\,445 \text{ examples (approximately).}}$$

Assignment 2. Let $\mathcal{X}$ denote a set of images of handwritten digits and let $\mathcal{Y} = \{0, 1, \ldots, 9\}$ be the corresponding set of digit labels. Consider a prediction rule $h\colon \mathcal{X} \times \mathcal{X} \to \mathcal{Y} \times \mathcal{Y}$ which, for each pair of images $(x_A, x_B) \in \mathcal{X} \times \mathcal{X}$, outputs a pair of predicted digit labels $h(x_A, x_B) = \big(h_A(x_A), h_B(x_B)\big) \in \mathcal{Y} \times \mathcal{Y}$. We measure the prediction accuracy of h using the absolute deviation between the sum of the true digit labels and the sum of the predicted digit labels. To this end, we define the loss function $\ell(y_A, y_B, \hat{y}_A, \hat{y}_B) = \big|y_A + y_B - \hat{y}_A - \hat{y}_B\big|$. The expected risk of the predictor h with respect to a joint probability density function $p(x_A, x_B, y_A, y_B)$ defined on $\mathcal{X} \times \mathcal{X} \times \mathcal{Y} \times \mathcal{Y}$ is given by $R(p, h) = \mathbb{E}_{(x_A, x_B, y_A, y_B) \sim p}\left[\ell\big(y_A, y_B, h_A(x_A), h_B(x_B)\big)\right]$. Since the distribution $p(x_A, x_B, y_A, y_B)$ is unknown, we estimate $R(p, h)$ using the test error

$$\widehat{R}(S_n, h) = \frac{1}{n}\sum_{j=1}^{n} \ell\big(y_{A,j}, y_{B,j}, h_A(x_{A,j}), h_B(x_{B,j})\big),$$

where $S_n = \big((x_{A,j}, x_{B,j}, y_{A,j}, y_{B,j}) \in \mathcal{X} \times \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \mid j = 1, \ldots, n\big)$ is a test set consisting of l i.i.d. samples drawn from the distribution $p(x_A, x_B, y_A, y_B)$.

a) What is the minimal number of test examples n that must be collected in order to guarantee that

$$R(p, h) \in \big(\widehat{R}(S_n, h) - \varepsilon,\ \widehat{R}(S_n, h) + \varepsilon\big)$$

with probability at least $1 - \delta$? Express n as a function of ε and δ.

b) Provided the number of test examples is $n = 10,000$, what is the probability that $R(p, h)$ is in the interval $\big(\widehat{R}(S_n, h) - 1,\ \widehat{R}(S_n, h) + 1\big)$?

Solution 2. Hoeffding's inequality.

$$\mathbb{P}\Big(\big|\widehat{R}(S_n, h) - R(p, h)\big| \geq \varepsilon\Big) \leq 2\exp\bigg(-\frac{2n\varepsilon^2}{(\ell_{max} - \ell_{min})^2}\bigg) = 2\exp\bigg(-\frac{2n\varepsilon^2}{18^2}\bigg).$$

(a) Minimal number of test examples. From the Hoefftind inequality we can derive that

$$\boxed{n = \frac{\log(2) - \log(\delta)}{2\varepsilon^2}(\ell_{max} - \ell_{min})^2 = \frac{\log(2) - \log(\delta)}{2\varepsilon^2}18^2}$$

(b) Probability for $n = 10{,}000$ and $\varepsilon = 1$. Using Hoeffding's inequality,

$$\mathbb{P}\Big(\big|\widehat{R}(S_n, h) - R(p, h)\big| \geq 1\Big) \leq 2\exp\bigg(-\frac{2 \cdot 10{,}000 \cdot 1^2}{18^2}\bigg) = 2\exp\bigg(-\frac{20{,}000}{324}\bigg).$$

Since $\frac{20{,}000}{324} \approx 61.73$,

$$\mathbb{P}\Big(\big|\widehat{R}(S_n, h) - R(p, h)\big| \geq 1\Big) \approx 3.1 \times 10^{-27}.$$

Therefore,

$$\boxed{\mathbb{P}\Big(R(p, h) \in \big(\widehat{R}(S_n, h) - 1,\ \widehat{R}(S_n, h) + 1\big)\Big) \geq 1 - 2\exp\bigg(-\frac{20{,}000}{324}\bigg) \approx 1 - 3.1 \times 10^{-27}.}$$